

Exp 13

→ Write a program to define multiple macros to perform arithmetic functions.

```
#include <stdio.h>
```

```
#define ADD(a,b) ((a)+(b))
```

```
#define SUBTRACT(a,b) ((a)-(b))
```

```
#define MULTIPLY(a,b) ((a)*(b))
```

```
#define DIVIDE(a,b) ((a)/(b))
```

```
int main () {
```

```
    int x = 20;
```

```
    int y = 5;
```

```
    int result;
```

```
    printf (" --- C Program Demonstrating Arithmetic macros -\n");
```

```
    printf (" Variables: x = %d , y = %d\n\n", x, y);
```

```
    result = ADD(x, y);
```

```
    printf (" ADD(%d, %d) -> Result : %d\n", x, y, result);
```

```
    result = SUBTRACT(x, y);
```

```
    printf (" SUBTRACT(%d, %d) -> Result: %d\n", x, y, result);
```

```
result = MULTIPLY(x, y);  
printf("MULTIPLY(%d, %d) -> Result: %d\n", x, y, result);
```

```
result = MULTIPLY(x+1, y+1);  
printf("MULTIPLY(x+1, y+1) -> Result: %d (21*6)\n", result);
```

```
result = DIVIDE(x, y);  
printf("DIVIDE(%d, %d) -> Result: %d\n", x, y, result);
```

```
result = DIVIDE(x, 3);  
printf("DIVIDE(%d, %d) -> Result: %d\n", x, 3, result);
```

```
return 0;  
}
```


